

Data Science Applications

Assignment Semester 2 2026





Preamble

The main purpose of the assignment from your perspective is to help you to:

- consider the business environment in which a problem is to be solved;
- apply data science techniques to solve a business problem; and
- communicate the outcomes of your analysis to business stakeholders.

These skills will also help you pass the end of semester assessment and perform well in the workplace.

The specific skills that are being developed and assessed in the assignment are the ability to¹:

- evaluate how well data describes business activity;
- develop solutions to a range of classification problems using GLMs, tree-based models, ensembling and neural networks;
- evaluate solutions produced by classification models;
- perform k-means and hierarchical clustering;
- evaluate a clustering algorithm using internal, external, and manual validation;
- apply each step in the natural language processing pipeline to solve a variety of business problems;
- evaluate the outcomes of natural language processing models;
- implement strategies for gaining stakeholder support for data science projects;
- communicate relevant points in language appropriate to the audience, in a logical and coherent manner; and
- meet business standards for presentation of work, both modelling and written materials.

The assignment requires you to create a set of sensible **assumptions** or parameters. You need to demonstrate *how* you derived your assumptions or model parameters. It is important that you describe what you did as the marker(s) will want to understand if you are able to apply knowledge to the specific situation described in this assignment. We are also looking for you to demonstrate that you can deal with uncertainty in a reasonable way.

The assignment requires you to build and/or use a **model**. A key actuarial skill is to obtain a grasp of the qualitative nature of outputs from models and describe them. This assignment is designed to test your ability to explain your model(s) and their outputs to a non-technical audience.

¹ The skills listed here are learning objectives from the subject's syllabus, apart from the last two skills on the list which are assessable in every subject. This assignment does not cover every component of the learning objectives listed above.



There may not be a single correct answer to the question(s) posed. Ensure you have adequately demonstrated your steps, assumptions, reasoning and checks to the marker(s) so that your answer can be considered in context.

Marking guide

A percentage mark of at least 60% is required to achieve at least a pass grade for this assignment, which is consistent across all the subjects' assignments. This is only an indicative pass mark and the final pass mark for the subject overall may be different.

This assignment represents 50% of the available marks for the Data Science Applications subject². Your assignment mark will be combined with your exam mark to determine your overall result for the subject.

It is anticipated that you will spend at least 50 hours to complete the assignment. In past semesters, some students have spent significantly more time than this, particularly those students who aim for a grade of Above Pass Level or Significantly Above Pass Level.

A detailed rubric is provided with the assignment question and will be used by the markers to assess your performance. The rubric has been posted on the Assignments page of Canvas to guide you as to what is required to achieve full marks for each part of the assignment. You should check that the components of your answer cover the items in the rubric.

You should use a clear structure in your work, whether written, coded, or recorded to make it easy for markers to find where you have responded to each of the rubric criteria.

Submission

Deadline

The deadline for submission is Tue 29 September, 12:00 pm.

Submit your assignment via the Assignments page in Canvas. If you experience technological issues when submitting your assignment, please send a copy of your assignment by email to education@actuaries.asn.au.

² For students completing the subject as a microcredential Certificate path, the assignment represents 100% of the available marks for the microcredential.



Penalties apply for late submissions (see section on 'Penalties'). You should anticipate potential delays by preparing and submitting your work in advance of the deadline.

Should circumstances arise that mean you cannot submit your assignment on time, you should contact education@actuaries.asn.au in advance of the deadline and apply for special consideration.

File format

The submitted documents must consist of four files:

- an ipynb (Jupyter notebook) file – containing your answers to all questions;
- an html copy of the notebook;
- a pdf file – containing just your assignment coversheet; and
- a md or txt log file of your AI conversations (where you used AI tools).

Files in other formats will not be marked. The naming convention for files is:

DSA 2026 S2 Assignment member ID.(file extension as appropriate) or Submit AI_LOGS.md for your AI conversation logs.

Please note that if you resubmit an assessment, Canvas automatically adds a suffix to the file name (such as '-1' for the first resubmission). You do not have to make any adjustment for this.

Coversheet

A coversheet for the assignment is provided on the Assignments page in Canvas. Complete this coversheet and save as a pdf file.

Ensure you are complying with the statements provided on the coversheet.

Word limit

Some questions in the assignment have a specific word or page limit. Markers will not read any part of your answer that exceeds this limit. Keep your word count or page count within any limits that are specified. The word count includes any text within tables, text boxes or images consisting primarily of text. The word count does not include:

- contents table or index; and
- references to sources used.



Keep in mind one of the key principles taught in the Communication, Modelling and Professionalism subject: always write as clearly and succinctly as possible, while still including enough information that will be useful for your audience. With that in mind, consider whether each word, sentence or paragraph you include in your assignment adds to or detracts from the message you are trying to convey. Importantly, know that 'more' is usually not 'best'.

Video

As part of this assignment, you are required to record a video presentation. Advice about how to record an effective video is provided in a video instruction document available on Canvas ('Assignment Video Production Instructions.pdf'). You should submit your video by following these steps:

- create a video recording using the naming convention 'DSA 2026 S2 Assignment member ID';
- use your video recording to create an 'unlisted' YouTube video (see the video instruction document)³; and
- insert your YouTube video URL as a hyperlink at the bottom of your Jupyter notebook.

Time limit

The video may have a specified time limit. Markers will not watch any part of your answer that exceeds this limit. Keep your presentation timing within any limits that are specified.

Jupyter notebook

The Jupyter notebook should use the assignment notebook template provided ('DSA_2026_S2_Assignment_template.ipynb'). Within the notebook you should:

- explain each step taken in your analysis in a text cell above your code; and
- evaluate and comment on the output from each step in a text cell below the output.

³ The video instruction document also provides advice for students who do not have access to YouTube due to their location.



Please note that, except where specified, there is no word limit for the comments in your notebook. However, markers will look more favourably on students who provide clear and succinct commentary, compared to those who provide no commentary or those who provide too much commentary, including those who repeat large sections of the subject materials in their comments. This latter approach makes it very difficult for a marker to assess your understanding of the step being taken or the output being produced.

Before submitting your assignment, you should create an html copy of the notebook by following the instructions and running the code contained in the notebook 'DSA_assignment_convert_to_html.ipynb'.

Plagiarism

By submitting your assignment, you are implicitly stating that the work is your own.

Remember that an important aspect of being a professional actuary is to always act with integrity. Committing plagiarism by copying another person's work or not properly referencing other sources (including artificial intelligence sources) used in your assignment is Inappropriate Educational Conduct under the Educational Conduct Policy.

The Education Strategy Committee has the power to investigate and apply penalties, before or after results are released.

Penalties

Late submissions

Penalties will be applied to late submissions without prior approval.

If you submit an assessment after the due date and time (whether that is the original due date or any extended due date you have been granted), the following penalties apply:

- within 24 hours of due date and time: 20% x maximum mark available (i.e. deduct 4 marks if a 20 mark assignment, deduct 10 marks if a 50 mark assignment);
- more than 24 hours (1 day) late: 100% x maximum mark available (i.e. assessment score = 0).



Incorrectly formatted submissions

There is no direct penalty if an assessment is submitted in a format with an incorrect file name or an incorrect format (e.g. submitted as a word document when a pdf document was required). However, you may be required to resubmit your work with the correct file format, particularly relevant to modelling or coding assignments.

If a submission does not include the correct identifier (member ID) in the file name, then it may take time to identify you as the student and you may be asked to resubmit your work with an appropriate identifier.

If either situation arises then this will probably cause you to submit late and hence incur the late submission penalties outlined above. Students should therefore follow all assessment instructions provided.

Feedback

Our approach to feedback is for students to receive general feedback and a sample assessment marked as 'Significantly above pass level'.

You should review the general feedback that is provided to all students as well as the sample assessment. After reviewing the general feedback, you should use the rubric to grade the sample assessment and your submission. This will help you to compare the assessments and identify areas where your submission could have been improved.

Our belief is that this active approach to studying will provide you with a deeper understanding of where you need to improve. This is the best way for you to learn about your areas of strength and weakness. We do not provide students with individual feedback on their assessments.

At the end of the semester, you will receive:

- a letter to indicate whether you have passed or failed the subject;
- if you have failed the subject, a breakdown of your grade for each assessment;
- general feedback to all students about assessment performance; and
- sample assessment(s) that were graded as 'Significantly above pass level'.



Assignment Context

You are a data science consulting actuary engaged by **Asclepius Therapeutics** ('Asclepius'), an Australian biotechnology company dual-listed on the ASX and NASDAQ. Asclepius was founded as a contract development and manufacturing organisation (CDMO) specialising in biologics and has since expanded into novel therapeutics and gene therapies. Its commercial portfolio generates approximately \$800 million in annual revenue from contract manufacturing services for third-party biologics, but growth in this segment has stalled over the past two years due to overcapacity in the CDMO sector and pricing pressure from new entrants in Asia-Pacific.

Asclepius has a pipeline of four clinical programs in various stages of development:

- **ASC-101** (Phase 3) — a first-in-class monoclonal antibody targeting Factor XIIa for the prevention of hereditary angioedema (HAE) attacks. The Phase 3 pivotal trial has enrolled approximately 300 patients across 60 sites. Topline results expected in Q1 2027.
- **ASC-204** (Phase 3) — an anti-inflammatory biologic targeting IL-6 for reduction of major adverse cardiovascular events (MACE) in patients with chronic kidney disease (CKD). This is a large cardiovascular outcomes trial (CVOT) with approximately 2,500 patients enrolled and is the most expensive program in the company's pipeline. Topline results expected in Q4 2027.
- **ASC-370** (Phase 2) — an AAV-based gene therapy for severe haemophilia B. The Phase 2 dose-confirmation study has enrolled 45 patients. Interim efficacy data expected in Q2 2027.
- **ASC-415** (Phase 1) — an oral small molecule inhibiting ferroportin for sickle cell disease. Early-stage; no near-term readouts.

Asclepius currently has a market capitalisation of approximately \$6 billion and employs around 3,000 staff across its commercial operations and R&D. Eighteen months ago, the company acquired a respiratory-focused specialty pharmaceutical company for \$1.5 billion, aiming to diversify its revenue base. The integration has been slower than expected: the acquired product portfolio has lost market share to generics, and the board is now considering a significant impairment of the acquired intangible assets. The company generates approximately \$300 million in annual operating cash flow from its CDMO business, but this is substantially offset by clinical development costs of approximately \$20 million per month across its four programs. Net cash on hand is approximately \$250 million.



Asclepius's share price has fallen approximately 30% over the past 12 months, driven by concerns about the acquisition's performance, slowing CDMO revenue growth, and market uncertainty about the Phase 3 readouts. Last quarter, a competitor's Phase 3 HAE prevention trial reported positive results, and Asclepius's share price dropped 12% in sympathy — the market interpreted the competitor's success as reducing the commercial opportunity for ASC-101, despite ASC-101 targeting the disease through a different mechanism. Several other companies are running trials in the same therapeutic areas as Asclepius, and the board has observed that competitor readouts can move Asclepius's price even when the underlying science is unrelated. This episode left the board feeling unprepared and wanting a more rigorous, data-driven approach to anticipating market reactions.

The company's Chief Financial Officer (CFO) is preparing to raise approximately \$400 million in capital, primarily to fund the ASC-204 cardiovascular outcomes trial through to completion and regulatory submission, and to advance ASC-370 into Phase 3 if the interim data supports it. The board is weighing several funding options — including an equity raise, convertible notes, and royalty financing against the CDMO business — and the attractiveness of each depends on the expected trial outcomes and the share price at the time of the raise. An equity raise at the current depressed share price would mean issuing more shares for the same capital, significantly diluting existing shareholders. Conversely, a positive trial readout could lift the share price and allow the company to raise on much more favourable terms. However, the board is also aware that even positive clinical results do not always produce the share price increases that investors expect — particularly for large companies where success may already be partially priced in. The board needs to understand the range of likely outcomes before committing to a funding strategy and timeline.

Previously, the Investor Relations and Strategy team ('the IR team') relied on sell-side analyst reports and informal benchmarking to estimate likely share price impacts around trial readouts, but the board found these insufficiently rigorous for a decision of this magnitude. The IR team has since compiled a dataset from publicly available sources linking clinical trial records from ClinicalTrials.gov, FDA approval data, and an event study measuring abnormal share price returns around the dates when trial results were disclosed in SEC 8-K filings. The trial outcome classifications in the dataset were generated by an LLM pipeline that extracted and classified outcomes from the text of approximately 3,800 8-K filings using a dual-prompt consensus approach. A methodology document describing this classification process is provided with the dataset. The IR team does not have the data science expertise to validate this pipeline or build predictive models from the dataset.



The CFO has engaged you to help the IR team (who you can assume are commercially experienced but are not actuaries or data scientists) determine whether this dataset can be relied upon to inform the company's capital strategy. Specifically, the CFO wants you to assess the quality of the LLM-generated classifications, analyse how the market has historically reacted to trial announcements similar to Asclepius's upcoming readouts, investigate whether the magnitude of market reactions can be predicted from trial characteristics and filing text, and hand your work over to the senior actuary who will review it before it goes to the board.

The Dataset

The dataset provided contains:

- **Clinical trial records** from ClinicalTrials.gov for industry-sponsored interventional trials, including trial design characteristics, intervention descriptions, eligibility criteria text, conditions studied, sponsor details, and current status.
- **FDA approval data** from the Drugs@FDA database, including a drug-level matching table that classifies each drug name as FDA-approved or not, and derived features such as application type, years since approval, and the company's overall FDA approval track record.
- **8-K filing text** — the full text of approximately 3,800 SEC 8-K filings that disclosed clinical trial results, stored in the database.
- **LLM-classified trial outcomes** extracted from the 8-K filings using a dual-prompt consensus approach, with each result classified as positive, negative, mixed-positive, mixed-negative, inconclusive, or no result.
- **Event study results** measuring the abnormal share price returns observed around each 8-K filing date, calculated using a market model with a 120-day estimation window and a $[-1, +1]$ day event window.
- **An enriched analytical table** that links each event study result to its corresponding ClinicalTrials.gov registration, combining market reaction data (abnormal returns) with trial design features (enrollment, masking, allocation, endpoints, conditions, etc.) and a pre-computed disclosure lag variable. Where a trial announcement could not be linked to a specific trial registration, the trial design fields are null; the linkage metadata includes a match confidence indicator.
- **Filing-level summaries** that aggregate multi-drug filings to one row per filing, with a primary outcome classification and a mixed-outcome indicator.
- **Asclepius readout characteristics and illustrative filings** that include trial design, company context, and draft 8-K announcement text for Asclepius's three near-term readouts, written by their investor relations team



Note: Some 8-K filings disclose results for multiple clinical trials in a single filing. In these cases, the dataset contains one row per trial result, and the same filing (and therefore the same abnormal return) may appear multiple times in the event study results. The filing-level summary table provides one row per filing for analyses where filing-level deduplication is appropriate. You should consider how to handle multi-drug filings in your analysis.

The data dictionary for this dataset is set out as a supplementary file.

A **Getting Started notebook** is provided with this assignment. It introduces the key concepts you need (cumulative abnormal returns, clinical trial phases, 8-K filings), demonstrates how to load and join the database tables, and includes exploratory visualisations. A Supplementary Reading notebook is also provided. It extends the course material with worked examples, in Python, of techniques the assignment questions draw on. Both notebooks are required, and you should work through them before starting the assignment questions.

The use of AI tools

Actuarial practice increasingly means directing and taking responsibility for work you did not produce from scratch, including AI output. The principle underlying Practice Guideline 1 (PG1) is that a Member remains responsible for the judgements in their work, regardless of who, or what, produced it first. Meeting that responsibility means understanding a model before relying on it and judging whether its assumptions and methodology are appropriate for the purpose at hand.

You may use AI tools on this assignment; you are not required to. If you do, keep the actual conversations, including your prompts and the AI's replies, captured as you go, as described in *Recording your AI use* provided with this assignment.

Your saved conversations form part of your submission, and Question 5 asks you to account for how your work was produced. Conversations are difficult to recover after the event, so set up your record-keeping before you begin work.

You are not marked on how much or how little AI you use. Heavy use and light use can both score well, provided you can show your judgement shaped the outcome: a constraint you deliberately set, a result you checked, or an assumption you corrected rather than accepted by default. Full grading criteria are set out in the rubric.



What you have been given

The following accompany the database. The first two are required reading, and you should work through both before starting the questions.

- **Getting Started notebook** — introduces the key concepts (cumulative abnormal returns, clinical trial phases, 8-K filings), sets out how the tables relate and how to join them, works through an example filing, and explores the data.
- **Supplementary Reading notebook** — extends the course material with worked examples, in Python, of techniques these questions draw on.
- **Methodology document** — how the LLM classifications, the event study and the study–event linkage were produced.
- **Data dictionary** — every table and column in the database.
- **Recording your AI use** — how to record and submit the AI conversation logs that Question 5 requires.



Assignment Questions (Total: 100 Marks)

Answer Questions 1, 2, 3, and 4 in your Jupyter notebook using the assignment template provided. Share your response to Question 5 as a link in the notebook, following the Assignment Video Production Instructions and submit the AI usage log as a separate file.

1. Validate and improve the dataset (21 marks)

Before the CFO can rely on this dataset for capital planning, the LLM-generated classifications need to be evaluated for reliability, and the data you will use needs to be cleaned and its text fields vectorised.

- a. Assess the quality of the LLM-generated trial outcome classifications provided in the dataset. Your assessment should include two analyses:
- Using a random seed equal to your member ID, select a stratified random sample of at least 20 classified outcomes, ensuring the sample includes examples of each of the six documented outcome categories. Examine each sampled classification against the original 8-K filing text to assess whether the classification is accurate and whether the category boundaries (e.g. between positive and mixed-positive) are applied consistently.
 - Compare the LLM-classified outcomes to the sign of the return across the full dataset. Investigate cases where these disagree — for example, a trial classified as positive but associated with a negative abnormal return. For a selection of these disagreements, read the underlying 8-K filing to assess whether the disagreement reflects a classification error, market expectations, or other factors. These may be the same filings as your first sample or a fresh selection.

The methodology document provided with the dataset describes the approach used to generate the classifications; you should use this document to inform your evaluation.

(7 marks)



- b. Construct a systematic approach for flagging classifications in the dataset that are most likely to be unreliable, informed by your findings from Question 1a. Your approach should prioritise which classifications would benefit most from manual review, using patterns you observed. Your answer should include the results of applying your flagging approach to all classifications in `consensus_outcomes`, reporting how many and what proportion are flagged, broken down by outcome category. **(5 marks)**

- c. Clean the data you will use in this assignment and vectorise its text fields. You will likely return to this as you go: cleaning is judged on whether it serves the uses below, not on how exhaustive it is, and revisiting it once you can see how the features behave downstream is expected rather than rework.

The primary text source is the 8-K filing text in `filing_text`; you may also use trial-level text fields from `studies`. The filing text was extracted programmatically and provided without cleaning, so examine it before deciding what to do with it. Your vectorisation should include word or sentence embeddings with dimensionality reduction applied, and should avoid leakage (relative to the outcome you will later predict).

These features are used for the clustering in Question 2, where the full dataset is used because clustering is descriptive, and may inform your feature engineering in Question 3, where cross-validation applies. Consider how your approach should differ between those two contexts, and explain how you satisfied yourself that what you finally used is fit for both.

(5 marks)

- d. Summarise, in 500 words or less, what your assessment of the classifications and your flagging approach mean for the IR team, who may rely on this data to estimate market reactions to Asclepius's upcoming readouts. Address whether the classifications are fit to rely on, where they are weakest and whether that weakness matters for those estimates. Your answer should be communicated using language suitable for sharing with the IR team at Asclepius. **(4 marks)**



2. Identify comparable trials (20 marks)

Asclepius's board needs to understand how the market has historically reacted to trials similar to its own upcoming readouts. Your task is to segment the clinical trials in the dataset into groupings that can be used to identify which historical trials are the most relevant comparables for ASC-101 (HAE prevention), ASC-204 (cardiovascular outcomes in CKD), and ASC-370 (gene therapy for haemophilia B).

A trial is a useful comparable here if what happened to it tells the board something about what might happen to Asclepius's asset. Similarity might rest on disease area, trial design, stage, sponsor, or some combination of these — and whether trials that look similar actually move the market in similar ways is for you to establish rather than assume. Note that the number of directly comparable historical trials differs substantially across the three assets.

You should work with the trials for which event study data is available, so that the clustering outputs can be linked to observed market reactions in Question 4. The enriched analytical table in the dataset provides trial design features (enrollment, masking, allocation, etc.) alongside market reaction data for this purpose. At this stage, you are performing descriptive analytics, and therefore you should not partition the data.

- a. Apply a clustering algorithm using a combination of the vectorised text features from Question 1c and structured trial characteristics to segment trials into groupings. You should justify your choice of features, clustering algorithm, distance measure, and number of clusters, comparing your choices to alternatives. Use internal validation and the business context to support your justifications. Discuss the implications of your cluster size distribution. **(8 marks)**
- b. Apply discriminator modelling to identify, for each cluster, the ten keywords that most distinguish it from the others. Construct a TF-IDF vectorisation of the filing text, and train a supervised classifier to predict your cluster labels (from Question 2a) using only those TF-IDF features, without dimensionality reduction, so that individual terms remain identifiable. You should use a random forest algorithm with default hyperparameters and extract variable importance. You may need to adapt the standard feature importance approach to obtain per-cluster results (e.g. by fitting separate one-vs-rest classifiers). You are not required to assess the performance of the random forest model. **(4 marks)**
- c. Examine the clustering outputs using manual validation. Your manual validation should use approaches beyond the discriminator keywords from Question 2b. **(4 marks)**



- d. Summarise, in 500 words or less, the results of your cluster analysis. Describe the trial groupings you have identified, how much of the variation in observed market reactions they account for, and whether that is enough for the IR team to rely on them for that purpose. Your answer should be communicated using language suitable for sharing with the IR team at Asclepius. **(4 marks)**

3. Predict the magnitude of market reactions **(27 marks)**

The CFO has asked whether the size of the market reaction to a trial readout can be predicted before it happens. Which way the market moves is largely determined by the trial outcome itself, which you already have. How far it moves is not — whether a negative result means a 5% decline or a 50% crash. Narrowing that range is what the CFO needs when deciding on a capital raising strategy in Question 4. Note that the filing text for the Asclepius readouts is illustrative rather than real, so part of what Question 3 establishes is how much of your predictive power depends on filing text at all.

Research has shown that trial and company characteristics predict some of this variation. The dataset also holds the full text of the 8-K filings, so the question is whether the language of a disclosure adds anything beyond them.

Model the signed CAR value (`car_3day`), such as -25% or $+8\%$, not its absolute value.

- a. Construct a feature set for predicting the magnitude of market reactions, drawing on at least three distinct groups:
1. Trial and outcome features from the enriched analytical table.
 2. Text-based features extracted from the 8-K filing text, using at least two different text representation approaches.
 3. Company or market context features. The `stock_prices`, `event_fda_features` and `event_study_enriched` tables contain sufficient data for this group.

Every feature you construct must be observable before the market reaction it is used to predict. For each group, explain the choices that were yours to make: what you constructed and why, how you time-matched each feature to the filing date, how you treated filings covering more than one trial, and what you did where data was missing. **(7 marks)**



- b. Construct at least two predictive models from different model families to predict the signed CAR for each event. Your answer should include:
- A justification for each model family, explaining why it suits this task given the characteristics of the data, and why you did not use an alternative you considered.
 - Appropriate cross-validation to evaluate out-of-sample performance.
 - Standard regression metrics, and at least one evaluation showing whether the predictions have practical discriminatory power — for example, whether the events your model ranks worst do in fact have worse returns than those it ranks best.
 - An assessment of how much each feature group contributes: whether the text features add predictive value beyond the trial, outcome and company features, and by how much.
 - A discussion of any features that might constitute data leakage — features that appear predictive only because they are observed at the same time as, or after, the outcome — how you identified them, and what difference they made. **(10 marks)**
- c. Examine whether the predictability of market reactions is symmetric across outcome groups — specifically, whether the magnitude of negative reactions (crashes) is more predictable than the magnitude of positive reactions. Your examination should use model performance metrics computed within each outcome group, and include possible explanations. You may wish to consider concepts such as information asymmetry in bad-news disclosures, loss aversion, or the relative diversity of success versus failure narratives. **(6 marks)**
- d. Summarise, in 500 words or less, your findings on predicting the magnitude of market reactions. Your summary should address: how much of the variation in market reactions can your model explain? Which feature groups contribute most? What are the key limitations, and how confident should the CFO be in using these predictions to narrow the range of scenarios in Question 4? Your answer should be communicated using language suitable for sharing with the IR team at Asclepius. **(4 marks)**



4. Apply your analysis and prepare for deployment (17 marks)

The IR team has been following your work and wants to see what your pipeline produces before it takes anything further. To give it something to run on, the team has drafted announcement text for each of the three near-term readouts under a positive, a negative and a mixed outcome, supplied with the dataset. These are the team's own drafts rather than real filings — rough, and written by people who are not disclosure specialists but the team would rather test the approach now than wait for a readout. The IR team expects to take the results to the board.

- a. Estimate the range of likely market reactions to each of Asclepius's three near-term readouts, under positive, negative and mixed outcome scenarios, using your clustering from Question 2 and your magnitude model from Question 3.

Your answer should:

- (i) state which of the features you used in Questions 2 and 3 are available for a trial that has not yet reported, which are supplied, and which are neither;
- (ii) produce your estimates from the features you have, refitting or re-scoping your model as necessary, and state which you did and why;
- (iii) acknowledge what unavailable features may cost you;
- (iv) state the resulting range for each readout and scenario, and how much of its width is uncertainty about the trial outcome as opposed to a limit of your model; and
- (v) comment on the strength of the evidence behind each estimate — how well each readout is represented in your comparable set, and where the evidence base is thinnest.

If your model cannot separate the scenarios by more than its own run-to-run variability, say so and show the comparison. That is a valid and complete answer. **(10 marks)**

- b. Propose how the IR team should validate, monitor, and refresh your analytical pipeline after deployment, so that they can repeat this analysis for future readouts — both Asclepius's own and competitors'. Your answer should address: how the pipeline's outputs should be validated before the IR team relies on them; what should be monitored over time (e.g. model drift, changes in market behaviour, new trial types not represented in the training data); and what events should trigger a model refresh or rebuild. **(4 marks)**
- c. The IR team wants the board's support for using this approach on future readouts. Advise the team, in 300 words or less, on how to present it: what to put in front of the board and what to keep in the background, and how to respond if the board presses for a narrower range than the evidence allows. The IR team is commercially experienced but not technical. **(3 marks)**



5. Video presentation (15 marks)

Your analysis from Questions 1 to 4 will be reviewed by a senior actuary at Asclepius before the IR team relies on it.

Record a three-to-five-minute video, speaking to camera, handing your work over to that reviewer. The reviewing actuary is technically capable but has not seen your work before. Your handover must address, within the video:

- your opinion on whether the dataset and your analysis can be relied upon to inform the IR team's estimates of market reactions to Asclepius's upcoming readouts: what your analysis supports, and where it should not be relied upon;
- how the work was produced, including the nature and extent of any reliance on AI tools; and
- how you examined the work and satisfied yourself that it is sound.

Frame your account with reference to your responsibilities under PG1. Submit your AI conversation logs as described in 'Recording your AI use.docx'. If you completed the assignment without AI assistance, state this in the video and account for how you verified your own work.

Note that the account you give in the video and the AI logs must be a truthful account of your own work and must not misrepresent how the work was produced. **(15 marks)**

END OF ASSIGNMENT